



Vectors in Kinematics · Worksheet 1

Mechanics Chapter 8: Vectors in Kinematics. Answer all questions. Show your working clearly. Total: 66 marks. Time: 80 minutes.

Name _____

Class _____

Date _____

INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are three sections, one for each topic in Chapter 8. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- **i** and **j** are perpendicular unit vectors; in Section C, **i** is horizontal and **j** is vertically upwards. Take $g = 9.8 \text{ ms}^{-2}$. Distances are in metres and times in seconds unless stated.
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. [3].

Section	Marks available	Marks scored
A Position, velocity and acceleration vectors	19	
B Using calculus with vectors	21	
C Projectiles using vectors	21	
Roar-level challenge	5	
Total	66	

Section A Position, velocity and acceleration vectors

19 marks

1. A particle has velocity $(3\mathbf{i} - 4\mathbf{j}) \text{ ms}^{-1}$. Find its speed, and the angle its direction of motion makes with **i**. [2]

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2. A particle moves with constant velocity from the point with position vector $(2\mathbf{i} + \mathbf{j}) \text{ m}$ to the point with position vector $(8\mathbf{i} - 7\mathbf{j}) \text{ m}$ in 2 s. [3]
Find its velocity and its speed.

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3. The position vector of a particle at time t is $\mathbf{r} = (3t^2 - 2)\mathbf{i} + 5t\mathbf{j}$. [3]
Find the velocity of the particle, and its speed when $t = 2$.

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4. [$\mathbf{i}$ is due east and $\mathbf{j}$ is due north.] A ship starts at the point with position vector $(2\mathbf{i} + 3\mathbf{j})$ km and [4]
moves with constant velocity $(4\mathbf{i} - 2\mathbf{j})$ km h^{-1} .

- (a) Write down its position vector after t hours.
- (b) Find the time when it is due east of the origin, and its position then.
- (c) Find the bearing on which the ship is moving.

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$\mathbf{i}$ is due east and $\mathbf{j}$ is due north. $\mathbf{i}$ is due east and $\mathbf{j}$ is due north. $\mathbf{i}$ is due east and $\mathbf{j}$ is due north.

5. At time t hours, the position vectors of two ships A and B are **[$\mathbf{i}$ is due east and $\mathbf{j}$ is due north.]**

$$\mathbf{r}_A = (2 + 3t)\mathbf{i} + (1 + 4t)\mathbf{j} \text{ km}, \quad \mathbf{r}_B = (10 - t)\mathbf{i} + (2 + 3t)\mathbf{j} \text{ km}.$$

- (a) Find $\overrightarrow{AB}$ in terms of t . **[2]**

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- (b) Find the time at which B is due north of A . **[2]**

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- (c) Find the distance between the ships at this time. **[1]**

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- (d) Show that the distance d km between the ships satisfies $d^2 = 17t^2 - 66t + 65$. **[2]**

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(Total for Question 5 is 7 marks)



Section B Using calculus with vectors

21 marks

6. The position vector of a particle is $\mathbf{r} = t^3\mathbf{i} + 2t^2\mathbf{j}$. Find expressions for its velocity and acceleration. [2]

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7. The velocity of a particle is $\mathbf{v} = 2t\mathbf{i} + (3 - t^2)\mathbf{j}$. Find the acceleration when $t = 1$, and its magnitude. [3]

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8. The acceleration of a particle is $\mathbf{a} = 6t\mathbf{i} - 2\mathbf{j}$. When $t = 0$, its velocity is $(\mathbf{i} + \mathbf{j}) \text{ m s}^{-1}$. [3]
Find its velocity when $t = 2$.

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9. A particle starts at the origin with velocity $(3\mathbf{i} + 4\mathbf{j}) \text{ m s}^{-1}$ and moves with constant acceleration $(2\mathbf{i} - \mathbf{j}) \text{ m s}^{-2}$. [4]

- (a) Find the velocity after 4 s, and show that the particle is then moving parallel to $\mathbf{i}$.
(b) Find the position vector of the particle after 4 s.

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10. At time t seconds, the velocity of a particle P is $\mathbf{v} = (3t^2 - 12)\mathbf{i} + (2t - 5)\mathbf{j} \text{ m s}^{-1}$. When $t = 0$, P is at the point with position vector $(\mathbf{i} + 2\mathbf{j}) \text{ m}$.

(a) Find the acceleration of P when $t = 1$, and its magnitude. [2]

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(b) Find the time at which P is moving parallel to $\mathbf{j}$. [2]

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(c) Find the position vector of P at this time. [3]

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(d) Find the speed of P when $t = 3$. [2]

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(Total for Question 10 is 9 marks)



Section C Projectiles using vectors

21 marks

11. A particle is projected from O with velocity $(10\mathbf{i} + 5\mathbf{j}) \text{ m s}^{-1}$ and moves freely under gravity. Find its velocity after 1 s. [2]

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12. For the particle in Question 11, find its position vector at time t , and its position vector after 2 s. [3]

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13. A ball is projected from a point O on horizontal ground with velocity $(12\mathbf{i} + 14.7\mathbf{j}) \text{ m s}^{-1}$. [3]
Find the time taken to return to the ground and the horizontal distance travelled.

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14. A particle is projected from O with velocity $(15\mathbf{i} + 20\mathbf{j}) \text{ m s}^{-1}$. Find the greatest height reached, and the position vector of the particle at that moment. [4]

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15. A ball is kicked from a point O on horizontal ground with velocity $(14\mathbf{i} + 21\mathbf{j}) \text{ m s}^{-1}$. It passes through the point B with position vector $(28\mathbf{i} + h\mathbf{j}) \text{ m}$.

(a) Find h . [3]

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(b) Find the velocity of the ball at B . [2]

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(c) Find the angle the direction of motion makes with the horizontal at B , and state whether the ball is rising or falling. [2]

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(d) Find the speed of the ball at B . [1]

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(e) Find the greatest height of the ball above the ground. [1]

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(Total for Question 15 is 9 marks)



ROAR-LEVEL CHALLENGE

5 marks

16. Particle A starts at the origin and moves with constant velocity $(2\mathbf{i} + 3\mathbf{j}) \text{ ms}^{-1}$. At the same time, [5]
particle B starts at the point with position vector $(10\mathbf{i} - 4\mathbf{j}) \text{ m}$ and moves with constant velocity $(-3\mathbf{i} + 5\mathbf{j}) \text{ ms}^{-1}$.

Show that the particles collide, and find the position vector of the point where they collide.

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END OF WORKSHEET

Total: 66 marks